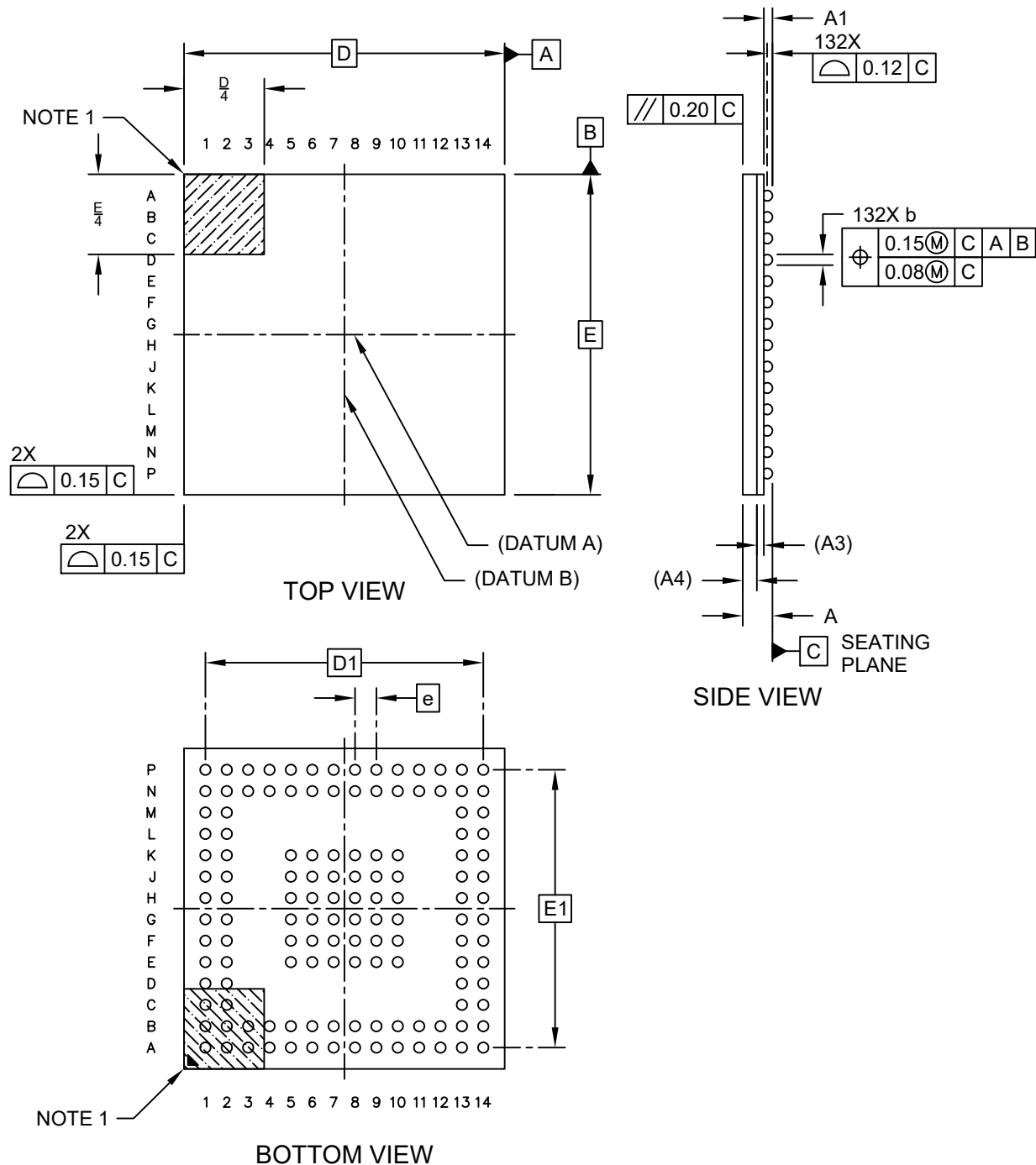


Packaging Diagrams and Parameters

132-Ball Thin Fine Pitch Ball Grid Array (AHA) - 12x12x1.2mm Body [TFBGA]

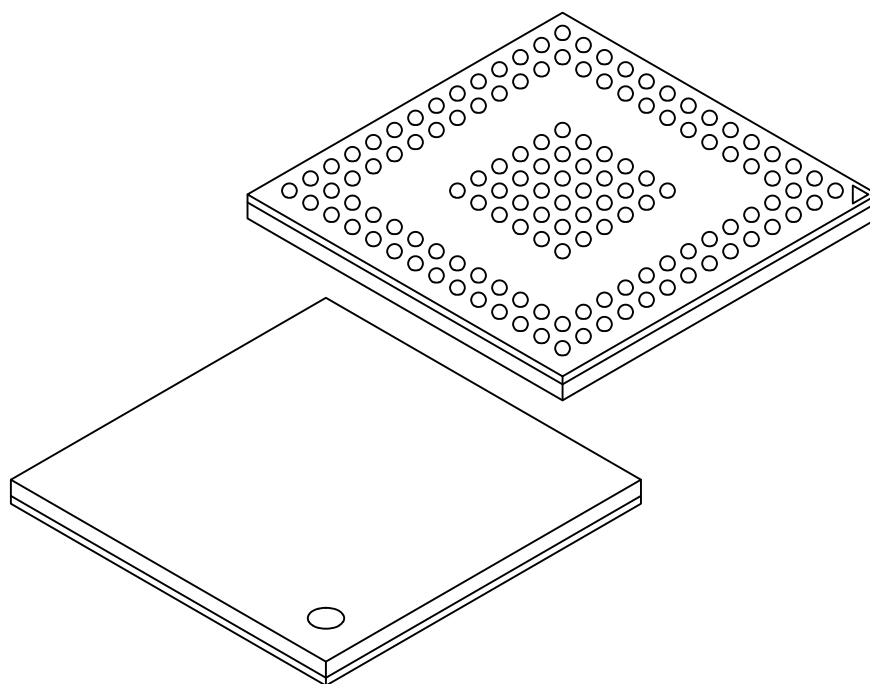
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Packaging Diagrams and Parameters

132-Ball Thin Fine Pitch Ball Grid Array (AHA) - 12x12x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		132		
Pitch	e		0.80 BSC		
Overall Height	A		-	-	1.20
Standoff	A1		0.27	0.32	0.37
Substrate Thickness	A3		0.26 REF		
Mold Cap Thickness	A4		0.53 REF		
Overall Length	D		12.00 BSC		
Overall Terminal Centers	D1		10.40 BSC		
Overall Width	E		12.00 BSC		
Overall Terminal Centers	E1		10.40 BSC		
Terminal Diameter	b		0.35	0.40	0.45

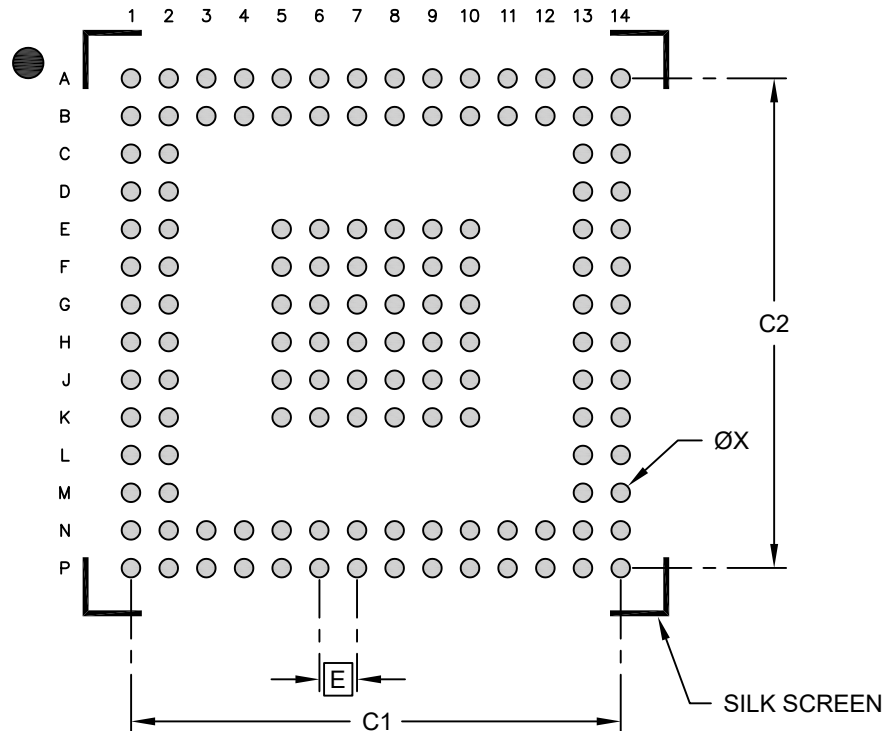
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

132-Ball Thin Fine Pitch Ball Grid Array (AHA) - 12x12x1.2mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Diameter	ØX		0.40	
Contact Pad Spacing	C1		10.40	
Contact Pad Spacing	C2		10.40	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-3193 Rev C